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Stock Price Prediction With Variational Mode Decomposition, Ecosystem-Based Optimization, and Radial Basis Function Models: Korea Composite Index Insights

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forecasting models. Concluding the historical data obtained from the KOSPI between January 2018 and January 2024, the current study tackles the enormous challenge of predicting stock prices. Various decomposition approaches were employed because financial time series are nonlinear. Variational Mode Decomposition (VMD) is selected to adaptively decompose intrinsic mode functions (IMFs) that are more effective in extracting the multiscale features of the original signals. Artificial Ecosystem-Based Optimization (AEO) is added as a metaheuristic search algorithm to guarantee the optimal setting of model parameters and improve generalization. The Radial Basis Function (RBF) network, the main predictive element, is especially suitable for financial forecasting applications due to its high nonlinear function approximation capability, quick convergence, and low computational cost. The suggested framework consistently outperforms the models because of its superior accuracy and stability. Its superiority in revealing nonlinear interdependencies and dynamic patterns in finance time series is attested to by its R^2 of 0.9955 on the KOSPI dataset. Generalizability of the proposed model was confirmed by applying it to a variety of financial

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AI: Artificial Intelligence

AO-GRU: Aquila Optimizer-Gated Recurrent Unit

Bi-GRU: Bidirectional Gated Recurrent Unit

CEEMD: Complementary Ensemble Empirical Mode Decomposition

CSI: China Securities Index

ELM: Extreme Learning Machine

GAN: Generative Adversarial Network

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min: Minimum

ML: Machine Learning

MLP: Multilayer Perceptron

MSGformer: Multi-Scale Graph Neural Networks With Transformer Encoders

OHLC: Open, High, Low, And Close

OLS: Ordinary Least Squares

ORELM: Outlier-Robust Extreme Learning Machine

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VMD: Variational Mode Decomposition

XGBoost: Extreme Gradient Boosting

XGBRF: Extreme Gradient Boosting Random Forest

XLSTM: Extended LSTM

b : Asterisk

w_k : Central Pulse

δ : Dirac Distribution

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N_v : Total Number Of Volume Nodes

UP : Upper Boundaries

$MaxIt$: Utmost Number Of Iterations

ω_i : Weight Correlation

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Ethics declarations

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